

2520030453

K . Siri Chandana

1. Virtual Address

```
siri_chandana@LAPTOP-MMV x + v
siri_chandana@LAPTOP-MMVUDEP:~$ mkdir -p ~/OS_Lab/VirtualAddress
siri_chandana@LAPTOP-MMVUDEP:~$ cd ~/OS_Lab/VirtualAddress
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/VirtualAddress$ nano virtual_address.c
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/VirtualAddress$ gcc virtual_address.c -o virtual_address
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/VirtualAddress$ ./virtual_address
Value of x = 10
Virtual address of x = 0x7ffc5fa91824
siri_chandana@LAPTOP-MMVUDEP:~/OS_Lab/VirtualAddress$ |
```

```
siri_chandana@LAPTOP-MMV x + v
GNU nano 8.7.1 virtual_address.c
#include <stdio.h>

int main() {
    int x = 10;

    printf("Value of x = %d\n", x);
    printf("Virtual address of x = %p\n", (void *)&x);

    return 0;
}

[ Read 10 lines ]
^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line  M-E Redo      M-G Copy
```

2. Process Spaces

```
siri_chandana@LAPTOP-MMV x + v
GNU nano 8.7.1 process_spaces.c *
#include <stdio.h>
#include <stdlib.h>

int global = 10;
int bss_variable;

int main() {

    int local = 20;

    int *heap = malloc(sizeof(int));
    *heap = 30;

    printf("Code address : %p\n", (void *)main);
    printf("Data address : %p\n", (void *)&global);
    printf("BSS address : %p\n", (void *)&bss_variable);
    printf("Heap address : %p\n", (void *)heap);
    printf("Stack address : %p\n", (void *)&local);

    free(heap);

    return 0;
}

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/_ Go To Line M-E Redo      M-6 Copy

siri_chandana@LAPTOP-MMV x + v
siri_chandana@LAPTOP-MMV:~$ cd ~/OS_Lab
siri_chandana@LAPTOP-MMV:~/OS_Lab$ nano process_spaces.c
siri_chandana@LAPTOP-MMV:~/OS_Lab$ gcc process_spaces.c -o process_spaces
siri_chandana@LAPTOP-MMV:~/OS_Lab$ ./process_spaces
Code address : 0x5ddbafc2f1a9
Data address : 0x5ddbafc32010
BSS address : 0x5ddbafc32018
Heap address : 0x5ddb6394010
Stack address : 0x7ffcc558f56c
siri_chandana@LAPTOP-MMV:~/OS_Lab$ |
```

3. Demand Paging

```
siri_chandana@LAPTOP-MMV x + v
GNU nano 8.7.1 demand_paging.c *
#include <stdio.h>

int main() {

    int page_table[4] = {1, -1, 3, -1};

    int page;

    printf("Enter page number (0-3): ");
    scanf("%d", &page);

    if (page_table[page] == -1) {
        printf("Page Fault!\n");
        printf("Page must be loaded into memory.\n");
    }
    else {
        printf("Page is already in memory.\n");
        printf("Physical frame = %d\n", page_table[page]);
    }

    return 0;
}

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute  ^C Location  M-U Undo    M-A Set Mark
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify  ^/ Go To Line M-E Redo    M-G Copy
```

```
siri_chandana@LAPTOP-MMV x + v
siri_chandana@LAPTOP-MMVUEP:~$ cd ~/OS_Lab
siri_chandana@LAPTOP-MMVUEP:~/OS_Lab$ nano process_spaces.c
siri_chandana@LAPTOP-MMVUEP:~/OS_Lab$ gcc process_spaces.c -o process_spaces
siri_chandana@LAPTOP-MMVUEP:~/OS_Lab$ ./process_spaces
Code address : 0x5ddbafc2f1a9
Data address : 0x5ddbafc32010
BSS address  : 0x5ddbafc32018
Heap address  : 0x5ddb6394010
Stack address : 0x7ffcc558f56c
siri_chandana@LAPTOP-MMVUEP:~/OS_Lab$ cd ~/OS_Lab
siri_chandana@LAPTOP-MMVUEP:~/OS_Lab$ nano demand_paging.c
siri_chandana@LAPTOP-MMVUEP:~/OS_Lab$ gcc demand_paging.c -o demand_paging
siri_chandana@LAPTOP-MMVUEP:~/OS_Lab$ ./demand_paging
Enter page number (0-3): 1
Page Fault!
Page must be loaded into memory.
siri_chandana@LAPTOP-MMVUEP:~/OS_Lab$ |
```

4. Address Translation

```

GNU nano 8.7.1 address_translation.c *
#include <stdio.h>

int main() {

    int page_table[] = {3, 5, 1, 7};
    int page_size = 100;

    int virtual_address = 250;

    int page = virtual_address / page_size;
    int offset = virtual_address % page_size;

    int frame = page_table[page];

    int physical_address = frame * page_size + offset;

    printf("Virtual Address = %d\n", virtual_address);
    printf("Page Number = %d\n", page);
    printf("Offset = %d\n", offset);
    printf("Frame Number = %d\n", frame);
    printf("Physical Address = %d\n", physical_address);

    return 0;
}

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute  ^C Location  M-U Undo     M-A Set Mark
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify  ^_/ Go To Line M-E Redo     M-6 Copy

```

```

siri_chandana@LAPTOP-MMV:~$ cd ~/OS_Lab
siri_chandana@LAPTOP-MMV:~/OS_Lab$ nano process_spaces.c
siri_chandana@LAPTOP-MMV:~/OS_Lab$ gcc process_spaces.c -o process_spaces
siri_chandana@LAPTOP-MMV:~/OS_Lab$ ./process_spaces
Code address : 0x5ddbafc2f1a9
Data address  : 0x5ddbafc32010
BSS address   : 0x5ddbafc32018
Heap address  : 0x5ddb6394010
Stack address : 0x7ffcc558f56c
siri_chandana@LAPTOP-MMV:~/OS_Lab$ cd ~/OS_Lab
siri_chandana@LAPTOP-MMV:~/OS_Lab$ nano demand_paging.c
siri_chandana@LAPTOP-MMV:~/OS_Lab$ gcc demand_paging.c -o demand_paging
siri_chandana@LAPTOP-MMV:~/OS_Lab$ ./demand_paging
Enter page number (0-3): 1
Page Fault!
Page must be loaded into memory.
siri_chandana@LAPTOP-MMV:~/OS_Lab$ cd ~/OS_Lab
siri_chandana@LAPTOP-MMV:~/OS_Lab$ nano address_translation.c
siri_chandana@LAPTOP-MMV:~/OS_Lab$ gcc address_translation.c -o address_translation
siri_chandana@LAPTOP-MMV:~/OS_Lab$ ./address_translation
Virtual Address = 250
Page Number     = 2
Offset          = 50
Frame Number    = 1
Physical Address = 150
siri_chandana@LAPTOP-MMV:~/OS_Lab$

```

5. Dynamic memory allocation

```

dynamic_memory_allocation.c:11:9: note: include '<stdio.h>' or provide a declaration of 'printf'
dynamic_memory_allocation.c:20:9: warning: incompatible implicit declaration of built-in function 'printf' [-Wbuiltin-declaration-mismatch]
    20 |         printf("%d ", p[i]);
        |         ^~~~~~
dynamic_memory_allocation.c:20:9: note: include '<stdio.h>' or provide a declaration of 'printf'
dynamic_memory_allocation.c:23:5: warning: incompatible implicit declaration of built-in function 'printf' [-Wbuiltin-declaration-mismatch]
    23 |         printf("\n");
        |         ^~~~~~
dynamic_memory_allocation.c:23:5: note: include '<stdio.h>' or provide a declaration of 'printf'
siri_chandana@LAPTOP-MMV:~/OS_Lab$ nano dynamic_memory_allocation.c
siri_chandana@LAPTOP-MMV:~/OS_Lab$ gcc dynamic_memory_allocation.c -o dynamic_memory_allocation
siri_chandana@LAPTOP-MMV:~/OS_Lab$ ./dynamic_memory_allocation
10 20 30 40 50
siri_chandana@LAPTOP-MMV:~/OS_Lab$

```

6. Segmentation fault

```
siri_chandana@LAPTOP-MMV × + v
GNU nano 8.7.1 segmentation_fault.c *
#include <stdio.h>

int main() {
    int *p = NULL;

    printf("Before accessing memory\n");

    *p = 100;

    printf("After accessing memory\n");

    return 0;
}

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute  ^C Location  M-U Undo     M-A Set Mark
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify  ^_ Go To Line M-E Redo     M-6 Copy

siri_chandana@LAPTOP-MMV × + v
siri_chandana@LAPTOP-MMV:~/OS_Lab$ cd ~/OS_Lab
siri_chandana@LAPTOP-MMV:~/OS_Lab$ nano segmentation_fault.c
siri_chandana@LAPTOP-MMV:~/OS_Lab$ gcc sengmentation_fault.c -o shell
cc1: fatal error: sengmentation_fault.c: No such file or directory
compilation terminated.
siri_chandana@LAPTOP-MMV:~/OS_Lab$ gcc segmentation_fault.c -o shell
siri_chandana@LAPTOP-MMV:~/OS_Lab$ ./shell
Before accessing memory
Segmentation fault ./shell
siri_chandana@LAPTOP-MMV:~/OS_Lab$ |
```